XRPL Validator — Daily Engineering Report · 2026-08-06

Period: 2026-08-06 00:00 to 24:00 US Eastern (2026-08-06 04:00 to 2026-08-07 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(probe): a dropped fetch is not a divergence — withhold the verdict** (`cae6b85` , xrpl-validator) Both probes conflated "the network dropped this fetch" with "this object does not exist", so an incomplete pre-state replayed into a plausible, well-formed divergence. #106099077 is what prompted this: the scout filed two `tecPATH_DRY vs tesSUCCESS` circular-arb Payments on the FFI leg, with a triage note blaming the offer-book directory walk and `succ` /overlay merge. Re-running the same fixture on the same binary — …
- **fix(ledger): a hydrated maker with no line for what it sells holds ZERO** (`44d7a1c` , xrpl-validator) #106096771 354/354 (was 352/354) and #106101763 224/224 (was 223/224) — three divergences across two independent ledgers, one root cause. Both came from the 2026-08-06 fresh scout batch, and #106101763 was never looked at while making the change: the first evidence this run that a fix generalises rather than fitting the ledger it came from.
- **fix(xrpl-ledger): auth is not an interest in a trust line** (`f3f8b98` , xrpl-validator) rippled's bLowReserveSet/bHighReserveSet (SetTrust.cpp:611-620) are exactly quality-in, quality-out, NoRipple-vs-DefaultRipple, freeze, limit and balance. We also tested lsfLowAuth/lsfHighAuth, so a single authorisation bit pinned a zero-balance, zero-limit line open forever.
- **feat(xrpl-ledger): DX_SEL — measure strand selection against rippled's model** (`dbb9110` , xrpl-validator) Settles what rippled actually does, and relocates the defect.
- **fix(xrpl-ledger): the bridge strand's upper bound must not charge the transfer fee** (`005ad9a` , xrpl-validator) rippled states the rule outright in adjustQualityWithFees (BookStep.cpp): "It is important that `qualityUpperBound` is an upper bound on the quality (it is used to ignore strands whose quality cannot meet a minimum threshold). When calculating quality assume no fee is charged, or the estimate will no longer be an upper bound." It therefore returns ofrQ UNCHANGED for a CLOB tip and for a multi-path AMM, and even in …
- **fix(xrpl-ledger): strand selection — upper-bound order with fall-through** (`03c2cb9` , xrpl-validator) The judge and the fall-through are ONE mechanism, which is why building the judge alone failed twice (recorded 8 v

17: it rejects the leading candidate and then crosses nothing).

- **fix(xrpl-ledger): a rejected candidate's offer removals must survive its rollback** (`a836f94`, xrpl-validator) Hunted for a ledger exercising the ofrsToRm deviation noted in 03c2cb9. There isn't one: DX_RM swept all 144 staged fixtures (batch2, newfinds, freshfinds, corpus) and reported ZERO cases where a rejected candidate's rollback discarded a reap. The one hit, #105912454, is the benign kind -- offers the fill CONSUMED, which rippled discards too, since it drops the failed strand's sandbox and carries out only ofrsToRm.

Independence program

No independence record for UTC day 2026-08-06; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Auto-Expiring Trust Line Authorization** (compliance) Add an optional sfAuthExpiration (UInt32 ledger close-time) to RippleState, settable by the trust-line holder that granted authorization via a new tfS
- **Domain-Scoped Deposit Authorization** (compliance) Add sfDepositAuthDomain (Hash256, a PermissionedDomain keylet) to AccountRoot, settable via AccountSet when asfDepositAuth is active. The existing dep
- **Oracle-Gated Payment Channel Claims** (payments) Add optional sfOracleConditionID (Hash256, a PriceOracle keylet), sfConditionThreshold, and sfConditionOp (UInt8: GTE/LTE) to the PayChannel object, s

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-06 05:44	—	xrpl-validator	`cae6b85`	fix(probe): a dropped fetch is not a divergence — withhold the verdict
08-06 13:46	—	xrpl-validator	`44d7a1c`	fix(ledger): a hydrated maker with no line for what it sells holds ZERO
08-06 18:47	—	xrpl-validator	`f3f8b98`	fix(xrpl-ledger): auth is not an interest in a trust line
08-06 21:48	—	xrpl-validator	`dbb9110`	feat(xrpl-ledger): DX_SEL — measure strand selection against rippled's model
08-06 22:41	—	xrpl-validator	`005ad9a`	fix(xrpl-ledger): the bridge strand's upper bound must not charge the transfer fee
08-06 23:52	—	xrpl-validator	`03c2cb9`	fix(xrpl-ledger): strand selection — upper-bound order with fall-through
08-07 01:40	—	xrpl-validator	`a836f94`	fix(xrpl-ledger): a rejected candidate's offer removals must survive its rollback

Times in UTC. A commit appears on the day it was committed to main or merged into it.